

DTS 204.2

Data Management I

Lecture Notes — Expanded to Match the Course Scheme

Godswill

How this document is organized. Everything Godswill wrote in the original notebook is transcribed here exactly, under the topic it belongs to. Green “Added context” boxes add small clarifications to existing notes. Amber “From the course scheme” boxes are full new sections, written to cover topics listed in the Table of Content that the photographed pages hadn’t reached yet — nothing from the original notes has been removed or altered.

Contents

1	Course Outline	2
2	Population and Sample	3
2.1	Sampling Methods	3
2.2	The Central Limit Theorem	3
3	Asymptotic Statistics	5
4	Statistical Models and Random Sampling Distribution	6
5	Elementary Time Series Analysis	7
5.1	Real-Life Scenario	7
5.2	Components of Time Series (continued)	7
6	Index Numbers	9
6.1	Limitations of CPI	10
7	Demographic Measures	11
8	Estimators (Point and Interval) and Test of Hypothesis	12
9	Regression and Correlation	14
9.1	Regression and Correlation — Further Notes	14
10	Programming in Python (Computing Language)	15
11	Computation of Mean, Variance, and Correlation	16
12	Sorting and Ranking of Data	17
13	Data Step Processing	18
14	Preparing Data for Analysis	19
15	Evaluating Qualitative Data	20

16 Sample Size Estimation	21
17 Basic Statistical Computing in Regression Analysis and the Analysis of the Design of Experiment	22
18 Introduction to Monte Carlo	23

1 Course Outline

DTS 204.2 (Data Management I)

Table of Content

- Population and Samples
- Asymptotic Statistics Models
- Statistical Models and Random Sampling Distribution
- Elementary Time Series Analysis
- Index Numbers
- Demographic Measures
- Estimators (Points and Intervals) and test of Hypothesis concerning Population, mean and Proportion (1 and 2 Sample cases)
- Regression and Correlation
- Programming in Python Computing Language
- Computation of mean, variance, and correlation
- Sorting and Ranking of Data
- Data Step Processing
- Preparing Data for analysis
- Evaluating Qualitative Data
- Sample size estimation
- Basic statistical computing in regression analysis & the analysis of the design experiment
- Intro. to Montecarlo

2 Population and Sample

Subsets of population $\rightarrow x$ (sample)

$\bar{x}$ (Mean (Average) x) σ (Sample Standard Dev.)

Statistical Method (P)

- Extracted Information (P)
- Population $\longleftrightarrow$ Sample
- Parameter

Population $\longleftrightarrow$ Statistics, Sample $\longleftrightarrow$ Distribution [Distribution]

Importance of Statistics

- What is sampling dist.
- Distribution of sample statistics.

Added context: The note's arrows sketch out the core vocabulary pair of statistics: a **Population** is the entire group you actually care about (every student in a university, every transaction in a year, etc.), while a **Sample** is the subset you actually collect data on because measuring the whole population is expensive or impossible. A **Parameter** describes the population (e.g. the true population mean μ) and is usually unknown; a **Statistic** describes the sample (e.g. the sample mean $\bar{x}$) and is computed from data you actually have. We use sample statistics to *estimate* unknown population parameters — this is the whole logic behind “Estimators” later in Section 8.

2.1 Sampling Methods

Sampling method

Simplified Random Sample (single R.S) — Everyone [each unit] has an equal chance.

Stratified sampling method — Split population into groups to look normal, used to reduce sampling dist issues, mixed.

Clustering — Split into clusters.

Added context: Quick side-by-side of the three sampling methods the note lists:

- **Simple Random Sampling (SRS):** every individual in the population has an equal, independent chance of selection — like drawing names from a hat.
- **Stratified Sampling:** the population is first split into homogeneous sub-groups (“strata”) — e.g. by gender, faculty, or age band — and a random sample is drawn from *each* stratum. This guarantees representation across important sub-groups, which plain SRS doesn't guarantee.
- **Cluster Sampling:** the population is split into naturally occurring clusters (e.g. by neighbourhood or department); you randomly pick a handful of *whole clusters* and sample everyone (or nearly everyone) inside them. It's cheaper when the population is spread out geographically, at some cost to precision.

2.2 The Central Limit Theorem

For a very large population, **sampling dist.** tends to look normal, it [the distribution] is normal, using **CLT**.

Asymptotics Behaviour of sample size (n): what $n \rightarrow \infty$.

if variance is $n < 30$ (use T -test)

if variance is $n \geq 30$ (use Z -test)

Note: σ is unknown $\Rightarrow$ use T -test.

3 Asymptotic Statistics

From the course scheme (not in the original notes): **Asymptotic statistics** studies how the behaviour of statistical estimators (like the sample mean $\bar{x}$ or sample proportion $\hat{p}$) changes as the sample size n grows toward infinity. It answers questions like: “Does my estimator get closer and closer to the true parameter as I collect more data?” and “What shape does its sampling distribution approach?”

Three ideas the original notes already gesture toward (the $n \rightarrow \infty$ line and the T -test/ Z -test split) sit inside this broader topic:

- **Consistency:** An estimator is consistent if it converges (in probability) to the true parameter value as $n \rightarrow \infty$. For example, the sample mean $\bar{x}$ is a consistent estimator of the population mean μ .
- **The Central Limit Theorem (CLT):** As n grows large (commonly $n \geq 30$ is used as a rule of thumb, exactly as the notes state), the sampling distribution of $\bar{x}$ approaches a Normal distribution *regardless of the shape of the original population distribution*, with

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right).$$

This is precisely why the notes say “for a very large population, sampling dist. tends to look normal.”

- **Why the $n < 30$ vs $n \geq 30$ rule matters:** When n is small *and* the population standard deviation σ is unknown (as flagged in the note’s box), the sampling distribution of the standardized mean follows a t -distribution (heavier tails than Normal) rather than exactly a Normal distribution, so we use the T -test. Once n is large, the t -distribution converges to the standard Normal distribution, which is why the Z -test becomes an acceptable approximation for $n \geq 30$.

Asymptotic results are what justify using “large-sample” formulas and normal-based confidence intervals even when we don’t know the exact underlying population distribution — they describe what happens “in the limit” as data accumulates.

4 Statistical Models and Random Sampling Distribution

From the course scheme (not in the original notes): A **statistical model** is a mathematical representation of the process that is assumed to have generated the observed data — e.g. assuming exam scores are Normally distributed with some mean μ and variance σ^2 , or assuming a series of successes/failures follows a Binomial distribution. Building a statistical model means (1) choosing a plausible probability distribution family, and (2) estimating its unknown parameters from sample data.

A **sampling distribution** is the probability distribution of a given *statistic* (like $\bar{x}$ or $\hat{p}$) obtained from repeatedly drawing samples of the same size n from the population. It's the bridge between a single sample and general conclusions about the population:

Population $\xrightarrow{\text{repeated random sampling}}$ many possible samples of size n $\xrightarrow{\text{compute statistic each time}}$ Sampling distribution

Key facts used throughout the course:

- The **mean of the sampling distribution of $\bar{x}$** equals the population mean: $E(\bar{x}) = \mu$.
- The **standard deviation of the sampling distribution of $\bar{x}$** (called the *standard error*) is $\frac{\sigma}{\sqrt{n}}$ — it shrinks as n grows, which is the formal reason larger samples give more precise estimates.
- Per the Central Limit Theorem (Section 3), this sampling distribution is approximately Normal for large n , which is what makes Z - and T -tests (mentioned in Section 2) valid tools.

5 Elementary Time Series Analysis

5.1 Real-Life Scenario

Trend: Assume 1 dollar is ₦1345, but it might increase/decrease.

If, what time of the year does the dollar go up? Futures?

Seasonal.

Cyclical: Economic policies (Governance can affect the price).

Irregular: Time Service (Random/Unpredictable) — CBN placing ban on some MFB (relieve finance ban); fixed rainfall in P.H (Port Harcourt) — Rainfall in P.H has changed.

Climate has changed.

Irregular — P.H last experienced flooding till 2018.

Flooding from unexpected source.

Seasonal — The fluctuating increase of JAMB candidates.

JAMB is written yearly.

Added context: This “Real-life scenario” block is actually walking through the **four classic components of a time series** using the Naira–Dollar exchange rate and JAMB registration as running examples:

- **Trend** — the long-term overall direction of the series (the dollar generally strengthening or weakening over years).
- **Seasonal** — fluctuations that repeat at fixed, known intervals within a year (JAMB numbers spike every year around registration season).
- **Cyclical** — longer, less regular ups and downs, often tied to broader economic or policy cycles (CBN policy changes, economic booms/recessions), without a fixed calendar period.
- **Irregular (Random)** — unpredictable, one-off shocks that don’t fit a pattern (e.g. the 2018 P.H. flooding from an unexpected source).

5.2 Components of Time Series (continued)

Cyclical: Sharp decrease/increase in economy because of ...

Transport fare — price of fuel increase.

— When school resumes, especially after long break.

Seasonal.

Irregular:

— Fuel Subsidy, sudden fuel scarcity, independent markets go on strike.

Stratified sampling — equal chance of selection (simplified) Random Sample.

Time Service — Components of Time Service.

Time Service

Data collected overtime at regular intervals.

Four components of time service —

— Temperature

Z-test

— Lecturer testing average score of 300 students.

Bombs can clock if boys did better than girls — use Z-test (probable...).

$$\left(\frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \right) \geq \text{OR} < 0?$$

If $\bar{x}_1 - \bar{x}_2 = 0$, Z-test and Chi-Square test.

Real-life Uses of Z, T-test, and Chi-Square

Chi-Square test.

If (a proposed model) $\bar{x} \pm 30$ vs Z-test.

if $n \geq 30$ (use of homeostatic Alt., $n \leq 30$, use T-test).

Summary line from the notes: Z-test $\rightarrow n \geq 30$; T-test $\rightarrow n < 30$; Chi-Square $\rightarrow$ categorical/independence comparisons.

Added context: Putting the scattered fragments above together: the note is contrasting three hypothesis-testing tools by *when you'd reach for each one*.

- **Z-test** — used for comparing means when the sample is large ($n \geq 30$) or the population standard deviation σ is known (e.g. comparing boys' vs. girls' average test scores out of 300 students, as in the note's example).
- **T-test** — used for comparing means when the sample is small ($n < 30$) and σ is unknown, so the sample standard deviation must be used instead, which widens the distribution's tails (the *t*-distribution).
- **Chi-Square test** — used for categorical data: testing whether an observed frequency distribution differs from an expected one, or whether two categorical variables are independent.

Four components of Time Service at regular intervals.

— Temperature.

Data collected overtime at regular intervals.

Component	Description
Trend	Long term increase/decrease over time
Cyclical	Fluctuations at irregular time intervals
Seasonal	Regular pattern within 1 year
Random / Unpredictable	Sudden fluctuations / irregular

6 Index Numbers

Index Numbers shows the % change [across two] time periods.

x (Sample)

Price index shows: $\text{New Price} \times 100 / \text{Old Price}$.

$$\begin{aligned}\% \text{ change} &= \frac{\text{New Index} - \text{Old Index}}{100} \\ \text{Change} &= \text{Index} - 100\end{aligned}$$

Why

- Measures changes (Cost of Living getting higher)
- D.P. items have diff. prices
- Making comparisons
- Policy making (Govt. use to adjust Salary)

Types of P. Index

CPI — Consumer Price Index.

P.I — Index of Industrial Production.

Proxy Index — Index of Industrial Production, Change in total value (E).

Export Value Index — “”

Method of showing Price Index

$$\text{Unadjusted Price Index} = \left(\frac{\sum P_1 Q_1}{\sum P_0 Q_0} \right) \times 100$$

Laspeyres Index (Fixed price)

$$L = \frac{\sum P_1 Q_0}{\sum P_0 Q_0}$$

Paasche’s Index (Fixed quantity, $P = \dots$)

$$P = \frac{\sum P_1 Q_1}{\sum P_0 Q_1} \times 100$$

Added context: Reading the index-number formulas the note wrote: P_0, Q_0 denote the **price** and **quantity** in the *base* (old) period, while P_1, Q_1 denote them in the *current* (new) period.

- **Laspeyres Index** weights price changes using *old (base-period) quantities* Q_0 — it answers “what would it cost today to buy what we bought back then?”
- **Paasche’s Index** weights price changes using *current-period quantities* Q_1 — it answers “what does it cost today to buy what we buy now, versus what it would’ve cost back then?”

Laspeyres tends to overstate inflation (people substitute away from goods that get expensive, but the formula still weights them at old-period quantities), while Paasche tends to understate it. This substitution problem is exactly the “limitation” flagged in the notes below.

6.1 Limitations of CPI

Limitation of CPI = Doesn't capture Substitution Bias.

Quality Change

Added context: Two classic CPI limitations, spelled out:

- **Substitution bias:** CPI assumes people keep buying the exact same fixed “basket” of goods every period. In reality, when a good's price rises sharply, consumers substitute toward cheaper alternatives — so a fixed-basket index (like Laspeyres) overstates the true cost-of-living increase.
- **Quality change bias:** If a product's quality improves over time (e.g. a phone gets a better camera at the same price), part of a price increase may reflect that quality improvement rather than pure inflation, but a simple price index can't tell the two apart unless it's specifically adjusted (“hedonic” adjustment) for quality.

7 Demographic Measures

Demographic Measures — Statistical measures used to demonstrate size, structure, growth, etc., to measure population.

$$\text{Population Density (P.D)} = \frac{\text{Total no. of people}}{\text{Land area}}$$

$$\text{Growth rate} = \% \text{ change to measure Population}$$

Added context: A few more standard demographic measures that pair naturally with the two the notes give, useful for the exam:

- **Crude Birth Rate (CBR)** = $\frac{\text{No. of live births in a year}}{\text{Mid-year population}} \times 1000$
- **Crude Death Rate (CDR)** = $\frac{\text{No. of deaths in a year}}{\text{Mid-year population}} \times 1000$
- **Dependency Ratio** = $\frac{(\text{Population aged 0–14}) + (\text{Population aged 65+})}{\text{Population aged 15–64}} \times 100$ — measures the economic burden on the working-age population.

8 Estimators (Point and Interval) and Test of Hypothesis

From the course scheme (not in the original notes): This entire topic — “Estimators (Points and Intervals) and test of Hypothesis concerning Population, mean and Proportion (1 and 2 Sample cases)” — is on the scheme but wasn’t reached in the photographed notes, so it’s added here in full.

Point Estimation

A **point estimate** is a single value, computed from sample data, used as the “best guess” for an unknown population parameter. For example, the sample mean $\bar{x}$ is a point estimate of the population mean μ , and the sample proportion $\hat{p}$ is a point estimate of the population proportion p .

Interval Estimation

A **confidence interval (CI)** gives a range of plausible values for the parameter, together with a confidence level (commonly 95%). For the population mean, when σ is known (or n is large, tying back to Section 3):

$$\bar{x} \pm Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

When σ is unknown and n is small, the T -distribution is used instead:

$$\bar{x} \pm t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}}$$

For a population proportion:

$$\hat{p} \pm Z_{\alpha/2} \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Hypothesis Testing — General Steps

1. State the null hypothesis H_0 (no effect / no difference) and alternative hypothesis H_1 .
2. Choose a significance level α (commonly 0.05).
3. Select the appropriate test statistic (Z , T , or χ^2 — exactly the choice discussed in Section 5).
4. Compute the test statistic from the sample data.
5. Compare against the critical value (or use the p -value) and decide whether to reject H_0 .

One-Sample Tests

- **Testing a population mean:** $Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ (or the T version with s in place of σ).
- **Testing a population proportion:** $Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$

Two-Sample Tests

- **Comparing two means** (e.g. the note’s “boys vs. girls” scenario in Section 5):

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

- **Comparing two proportions:**

$$Z = \frac{(\hat{p}_1 - \hat{p}_2)}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

where $\hat{p}$ is the pooled proportion.

This is exactly where the note’s “Real-life Uses of Z, T-test and Chi-Square” fragment (Section 5) fits: the boys-vs-girls average score comparison is a two-sample mean test, and Chi-Square tests are typically used when comparing proportions/frequencies across categories rather than means.

9 Regression and Correlation

Gaussian produces an equation used to predict a relationship between Dependent (Y) & Independent Variable (X).

Output of Correlation — Single (or) Regression comes with eqn $\rightarrow Y = a + t$

Correlation — Symmetric $\rightarrow x \rightarrow y = y \rightarrow x$

Relation — Non-Symmetric $\rightarrow x \rightarrow y \neq y \rightarrow x$

$$r \rightarrow -1 \text{ to } 1, \quad k \rightarrow 0 ?$$

Added context: Reading these fragments together: a **correlation coefficient** r measures the strength and direction of a *linear* relationship between two quantitative variables, and always lies in the range $-1 \leq r \leq 1$ (matching the note's " $r \rightarrow -1$ to 1 "). $r = 1$ is perfect positive correlation, $r = -1$ is perfect negative correlation, and $r = 0$ (the note's " $k \rightarrow 0$ ") indicates no linear relationship.

Correlation is **symmetric**: the correlation between X and Y is the same as between Y and X — it doesn't distinguish cause from effect. **Regression**, by contrast, is **non-symmetric (directional)**: regressing Y on X (predicting Y from X) gives a different equation than regressing X on Y , because regression explicitly designates one variable as dependent and the other as independent.

9.1 Regression and Correlation — Further Notes

Regression and Correlation

Regression measures the strength & direction of linear types.

Correlation — between 2 Quantitative values.

Analysis — Measures the strength & Direction of linear types.

The most widely used is the ... (Pearson Correlation Coefficient)

Added context: The note trails off here, but the standard formula it's pointing toward is the **Pearson product-moment correlation coefficient**, the most widely used correlation measure for two continuous variables:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}}$$

And the corresponding simple linear regression equation predicting Y from X :

$$\hat{Y} = a + bX, \quad b = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad a = \bar{y} - b\bar{x}$$

This lines up with the note's earlier line "Regression comes with eqn $\rightarrow Y = a + t$ " — here b (slope) is the note's "t"/coefficient, and a is the intercept.

10 Programming in Python (Computing Language)

Programming in Python

Statistical Inference and modeling

MatLab

Key features of Python

- Variables
- Arithmetic Operation
- Functions(x)
- Loops

— Python code to compute mean.

— Calculating mean with functions.

- Variance, S.D

Added context: A short worked illustration of the mean/variance/standard-deviation Python code the note is pointing to (Section 10, “Computation of mean, variance, and correlation,” on the scheme):

```
data = [12, 15, 18, 22, 25]
n = len(data)

# Mean
mean = sum(data) / n

# Variance (sample variance, using n-1)
variance = sum((x - mean) ** 2 for x in data) / (n - 1)

# Standard Deviation
std_dev = variance ** 0.5

print("Mean:", mean)
print("Variance:", variance)
print("Standard Deviation:", std_dev)
```

In practice, Python’s `statistics` module (`statistics.mean()`, `statistics.variance()`, `statistics.stdev()`) or the `numpy/pandas` libraries (`numpy.mean()`, `numpy.var()`, `numpy.std()`, `DataFrame.corr()` for correlation) are used instead of writing loops by hand, which is what the note’s “Functions()” and “Loops” bullet points are gesturing toward as the two ways of organizing this computation.

11 Computation of Mean, Variance, and Correlation

From the course scheme (not in the original notes): This scheme item is directly continued from Section 9's Python notes, so the formulas are collected here for reference:

$$\text{Mean: } \bar{x} = \frac{\sum x_i}{n} \quad \text{Sample Variance: } s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1} \quad \text{Sample S.D.: } s = \sqrt{s^2}$$

$$\text{Correlation (Pearson's } r\text{): } r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}}$$

These three computations — mean, variance, and correlation — form the numerical backbone of nearly every other topic in this course: sampling distributions (Section 4) are built from repeated means, hypothesis tests (Section 8) are built from means and variances, and regression/correlation (Section 7) is built directly from the correlation formula above.

12 Sorting and Ranking of Data

From the course scheme (not in the original notes): **Sorting** means arranging a dataset's values in a defined order — ascending (smallest to largest) or descending (largest to smallest). It's usually the first step before computing things like the median, quartiles, or percentiles, since those all require the data to be in order.

Ranking assigns a position number (1st, 2nd, 3rd, ...) to each data point based on its sorted order. Ranking is central to *non-parametric* statistical methods (used when data doesn't meet the assumptions needed for *Z-/T-tests*), such as the Spearman rank correlation coefficient:

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

where d_i is the difference between the ranks of paired observations. This is the ranking-based counterpart to the Pearson correlation coefficient from Section 7, used when the relationship between two variables is monotonic but not necessarily linear, or when the data is ordinal rather than continuous.

Handling tied ranks: When two or more values are equal, they are typically assigned the *average* of the ranks they would have occupied (e.g. two values tied for 3rd and 4th place both receive rank 3.5).

13 Data Step Processing

From the course scheme (not in the original notes): “Data step processing” refers to the sequence of operations applied to raw data to transform it into a form ready for analysis — a term and workflow widely used in statistical packages such as SAS (where it is literally called the “DATA step”), but the same logic applies in Python/pandas workflows.

Typical stages in a data step:

1. **Reading in data** from a source file (CSV, Excel, database, etc.)
2. **Filtering** — keeping only the rows/observations relevant to the analysis.
3. **Transforming variables** — creating new columns from existing ones (e.g. computing age from a date of birth, or converting currency).
4. **Merging/joining** datasets from multiple sources on a common key (e.g. Student ID).
5. **Aggregating** — summarizing data by groups (e.g. average score per department).
6. **Outputting** the cleaned, structured dataset for the next stage of analysis (e.g. regression, hypothesis testing).

In Python, this workflow is most commonly carried out using the `pandas` library, e.g. `pd.read_csv()`, `df.dropna()`, `df.groupby()`, and `df.merge()`.

14 Preparing Data for Analysis

From the course scheme (not in the original notes): Before any statistical model or test can be trusted, the raw dataset generally needs to be prepared. Key steps:

- **Data Cleaning:** Identifying and correcting errors — typos, impossible values (e.g. a negative age), inconsistent category labels (“M”/“Male”/“male”).
- **Handling Missing Data:** Deciding whether to remove incomplete records, or fill them in (*imputation*) using the mean, median, or a model-based estimate.
- **Detecting Outliers:** Extreme values can distort statistics like the mean and correlation coefficient; common detection methods include the $1.5 \times \text{IQR}$ rule or checking for $|Z| > 3$.
- **Coding and Recoding Variables:** Converting categorical variables into numeric codes suitable for analysis (e.g. “Yes”/“No” $\rightarrow$ 1/0), sometimes called *dummy coding*.
- **Checking Assumptions:** Many classical tests (Z -test, T -test, Pearson correlation) assume the data is approximately Normally distributed and free of major outliers — this is worth verifying (e.g. with a histogram or a normality test) before proceeding.

This stage directly sets up the next scheme topic, “Evaluating Qualitative Data,” since categorical/qualitative variables usually need special handling during preparation.

15 Evaluating Qualitative Data

From the course scheme (not in the original notes): **Qualitative (categorical) data** records categories or labels rather than numeric measurements — e.g. gender, marital status, state of origin, or a Likert-scale survey response (Strongly Disagree ... Strongly Agree). Since the arithmetic operations used for numeric data (mean, variance, Pearson correlation) don't meaningfully apply to categories, qualitative data needs its own set of tools:

- **Frequency tables and bar charts:** Summarize how often each category occurs.
- **Mode:** The most frequently occurring category is the relevant “central tendency” measure for qualitative data (mean and median generally don't apply to unordered categories).
- **Chi-Square test of independence:** Tests whether two categorical variables are associated (e.g. is there a relationship between gender and course choice?) — this connects directly back to the Chi-Square test mentioned in Section 5.

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

where O_i is the observed frequency and E_i is the expected frequency in each category, under the assumption of independence.

- **Ordinal data** (categories with a natural order, like Likert scales) can additionally use rank-based methods, tying back to Section 11 (Sorting and Ranking).

16 Sample Size Estimation

From the course scheme (not in the original notes): Before collecting data, researchers often need to decide *how large* a sample should be to achieve a desired level of precision — too small a sample gives unreliable estimates; too large wastes time and resources.

For estimating a population mean (given a desired margin of error E and confidence level via $Z_{\alpha/2}$):

$$n = \left(\frac{Z_{\alpha/2} \cdot \sigma}{E} \right)^2$$

For estimating a population proportion:

$$n = \frac{Z_{\alpha/2}^2 \cdot p(1-p)}{E^2}$$

When no prior estimate of p is available, $p = 0.5$ is conventionally used, since it maximizes $p(1-p)$ and therefore gives the most conservative (largest) required sample size.

Key intuition: required sample size grows with the desired confidence level and the population's variability, and shrinks as the acceptable margin of error E is relaxed. This formula is also the practical link back to the “ $n \geq 30$ ” asymptotic rule of thumb in Section 3 — sample size estimation is how you decide, *before* collecting data, whether you're likely to land comfortably above or below that threshold.

17 Basic Statistical Computing in Regression Analysis and the Analysis of the Design of Experiment

From the course scheme (not in the original notes):

Regression Analysis — Computational Side

Building on Section 7’s regression formulas, “statistical computing” here refers to using software (Python, R, SPSS, Excel, etc.) rather than hand calculation to:

- Fit the regression line/coefficients (a , b from Section 7) to real datasets with many observations.
- Compute the **coefficient of determination** R^2 (the proportion of variance in Y explained by X):

$$R^2 = r^2$$

(the square of the Pearson correlation coefficient, for simple linear regression).

- Check regression assumptions — linearity, constant variance of residuals (homoscedasticity), and normally distributed residuals — typically via residual plots generated in software.

Design of Experiments (DOE)

The **design of experiments** is the branch of statistics concerned with planning studies so that the effect of one or more factors on an outcome (the *response variable*) can be estimated clearly, while controlling for other sources of variation.

Core ideas:

- **Treatments:** The different conditions being compared (e.g. different fertilizer types, different teaching methods).
- **Randomization:** Randomly assigning subjects/units to treatment groups, to prevent systematic bias.
- **Replication:** Repeating each treatment on multiple units, so that natural variability can be distinguished from a genuine treatment effect.
- **Blocking:** Grouping similar experimental units together (a “block”) before randomizing treatments within each block, to control for a known source of variability (e.g. testing across different classrooms).
- **Analysis of Variance (ANOVA):** The standard statistical computing tool for analyzing designed experiments with more than two groups — it tests whether the means of several groups are all equal, generalizing the two-sample T -test from Section 8 to more than two groups.

Regression and DOE are closely connected: both are ultimately about explaining variation in a response variable using one or more predictor/treatment variables, and both rely on the mean/variance computations from Section 10.

18 Introduction to Monte Carlo

From the course scheme (not in the original notes): Monte Carlo methods use repeated random sampling to obtain numerical results for problems that might be difficult or impossible to solve analytically. The name comes from the famous Monte Carlo casino, referencing the role of randomness.

Basic idea:

1. Define a domain of possible inputs.
2. Generate random inputs from a chosen probability distribution (using a computer's random number generator).
3. Perform a deterministic computation on each random input.
4. Aggregate (average) the results to approximate the quantity of interest.

Classic illustrative example — estimating π : Randomly scatter points inside a square that circumscribes a circle; the proportion of points that fall inside the circle approximates $\pi/4$, so:

$$\pi \approx 4 \times \frac{\text{points inside circle}}{\text{total points}}$$

Why it matters for this course: Monte Carlo simulation is widely used to:

- Approximate sampling distributions (Section 4) when the underlying theory is too complex to derive analytically — e.g. simulating thousands of random samples from a population and directly observing how $\bar{x}$ varies, rather than relying purely on the Central Limit Theorem.
- Estimate the power of a hypothesis test (Section 8), or check how robust a test is when its assumptions are violated.
- Perform sample size estimation (Section 14) by simulation when a closed-form formula isn't available.

This is naturally the capstone topic on the scheme — it ties together sampling (Section 2), the Central Limit Theorem (Section 3), and computing in Python (Section 9), by using simulated random samples, run in code, to approximate results about population parameters.